



SVKM's NMIMS
Mukesh Patel School of Technology Management & Engineering /
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Subject:- Web Programming

NMIMS Library Management System

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Introduction

Traditionally, library management relied heavily on manual record-keeping systems, where librarians maintained physical registers to log book issuances, returns, and catalog entries. While this method served its purpose in earlier times, it was inherently inefficient and error-prone. With the advent of the digital era, web-based library management systems have revolutionized this traditional approach. These systems act as centralized digital platforms that streamline operations and significantly enhance accessibility. This eliminates dependency on physical presence and enables 24/7 access to library services.

The NMIMS Library Management System represents a tailored solution designed specifically to meet the dynamic requirements of a modern academic institution. Unlike generic systems, it incorporates advanced logistical and user-centric features into a unified platform. At its core, the system provides comprehensive catalog management, seamless issue and return tracking, and intelligent search functionality that allows users to filter resources by author, genre, availability, or relevance.

Beyond basic operations, the platform introduces enhanced functionalities such as real-time seat occupancy tracking, enabling students to check study space availability before visiting the library. It may also integrate smart reservation systems for both books and seating, reducing uncertainty and improving resource utilization. A key distinguishing aspect of the NMIMS system is its emphasis on user experience and design sophistication. The interface is crafted with a premium aesthetic, supporting both Dark and Light modes to cater to different user preferences and lighting conditions.

From an administrative standpoint, the system provides powerful backend tools for librarians, including analytics dashboards, usage reports, and inventory management features. These tools help in data-driven decision-making, such as identifying high-demand resources, optimizing procurement strategies, and managing library space efficiently.

The development of this system was guided by the following key objectives:

- **Digitization & Automation:** To replace physical logs with a centralized database that automatically tracks every transaction.
- **Real-Time Visibility:** To provide students with a live view of book stock and room occupancy, reducing unnecessary physical visits.
- **Resource Optimization:** To maximize the utility of the library's physical assets, such as discussion rooms and PC terminals, through a structured booking system.
- **User Empowerment:** To give students a personalized dashboard where they can manage their own requests and track their academic activity independently.
- **Administrative Efficiency:** To streamline the administrative workflow by consolidating catalog management and request approvals into a single automated dashboard.

Requirement Analysis

The NMIMS Library Management System is designed to digitize and streamline traditional library operations by providing an efficient, secure, and user-friendly web-based platform. The requirement analysis identifies the functional and non-functional needs of the system, ensuring that it meets both administrative and student expectations. It serves as a foundation for system design and development by clearly outlining user requirements and system capabilities. Additionally, it helps in minimizing ambiguities and ensures that the final system aligns with institutional goals and user needs.

Functional Requirements

- Unified Authentication: Secure login and signup for students (SAP ID) and administrators (Admin ID) with support for OTP-based verification.
- Inventory Management: A centralized "Global Inventory" for admins to add, update stock, and manage the book catalog.
- Digital Seat & Room Booking: A live visualizer for library occupancy and a conflict-aware booking system for discussion rooms.
- Request Lifecycle: Tracking the status of book borrow requests and room bookings (Pending, Accepted, Rejected).
- Consolidated Student Profile: A single dashboard for students to view their history, active bookings, and profile details.

Non-Functional Requirements

- Responsive Layout: A site that adapts seamlessly to different screen sizes.
- Dual Mode Support: Full compatibility with both Dark and Light visual themes.
- Data Integrity: Preventing double-bookings and ensuring stock levels cannot be manipulated into invalid states.
- Micro-Animations: Providing subtle visual feedback for improved user interaction.
- Security: The system must ensure secure authentication, data protection, and role-based access control.
- Reliability: The system should maintain consistent performance with minimal downtime.
- Availability: The platform should be accessible 24/7 to allow students to access resources anytime.
- Compatibility: The system should work seamlessly across devices (desktop, tablet, mobile) and browsers.

Development Technologies

The project utilizes a robust and scalable "Full-Stack" technology stack:

1. Frontend: The user interface is built using HTML, CSS, and JavaScript, providing a responsive and interactive design. Advanced styling techniques and frameworks (such as Bootstrap or Tailwind CSS) enhance the visual appeal, including support for Dark and Light modes.
 - HTML5 & Semantic Tags: For structural integrity and SEO.
 - Vanilla CSS: Used for custom styling and implementing the dual-mode design system.
 - AngularJS (1.x): Employed for its powerful data-binding and single-page application (SPA) capabilities.
 - FontAwesome: For premium iconography.
2. Backend: The server-side logic is implemented using technologies such as Node.js, which handles request processing, authentication, and communication between the frontend and database.
 - Node.js: As the runtime environment.
 - Express.js: For routing and API development.
3. Database:
 - SQLite3: A lightweight, file-based SQL database for efficient data storage and portability.
4. Security & Utilities:
 - Bcrypt: For secondary encryption of user passwords.
 - Express-Session: For maintaining secure user sessions.
5. Development Tools:

Code editors such as Visual Studio Code and platforms like GitHub are used for coding, testing, and deployment purposes.
6. Web Browser Compatibility:

The system is designed to run smoothly on modern web browsers such as Chrome.

Implementation and Outputs

Core Modules

Authentication System

Features a dual-layered role-selection gateway, allowing users to choose between Student and Admin portals. Following selection, users interact with specific login and signup forms. A highlight of this module is the "Premium OTP Toast," which simulates a secure verification process with high-fidelity visual feedback, reinforcing the system's commitment to security.

The first screenshot shows the NMIMS Library login page. At the top, there is a header with the NMIMS Library logo and a 'Light' button. Below the header, a central box prompts the user to 'Please select your role to continue.' There are two options: 'Student' (with a graduation cap icon and the description 'Access books & seating') and 'Admin' (with a shield icon and the description 'Manage library system').

The second screenshot shows the login page after selecting a role. It features a 'Back to Roles' button at the top. Below it, there is a field for 'SAP ID (11 Digits)' with the value '70472400114'. Underneath is an 'OTP' field with the value '584728' and a 'Get OTP' button. A checkbox labeled 'Login via OTP' is checked. At the bottom, there is a large orange 'Login' button and a link that says 'Don't have an account? Sign Up'.

Library Floor Plan (Live Visualizer)

A "BookMyShow" style interactive seat visualizer (librarySeats.html) that provides a real-time representation of the library layout. The module uses dynamic random shifts to simulate live occupancy, giving the interface a "living" feel and allowing students to check availability before making the visit.



Room Booking Engine

It prevents past-date bookings and resolves time-slot conflicts by checking all existing reservations. The system utilizes accurate local-time (IST) calculations to ensure precise scheduling across different user sessions.

The screenshot shows the 'Book Discussion Room' booking interface. At the top, there's a header with the 'NMIMS Library' logo and navigation icons. Below the header, there's a central form for booking a room. The form includes a 'Select Room' dropdown menu showing 'Room A (Max: 4)', a 'Date' field with a calendar icon showing '12-04-2026', a 'Time Slot' dropdown menu showing '11:00-12:00', and a 'Number of Students (Max: 4)' input field with a value of '4'. A prominent orange 'Submit Request' button is at the bottom of the form.

Inventory Dashboard (inventory.html)

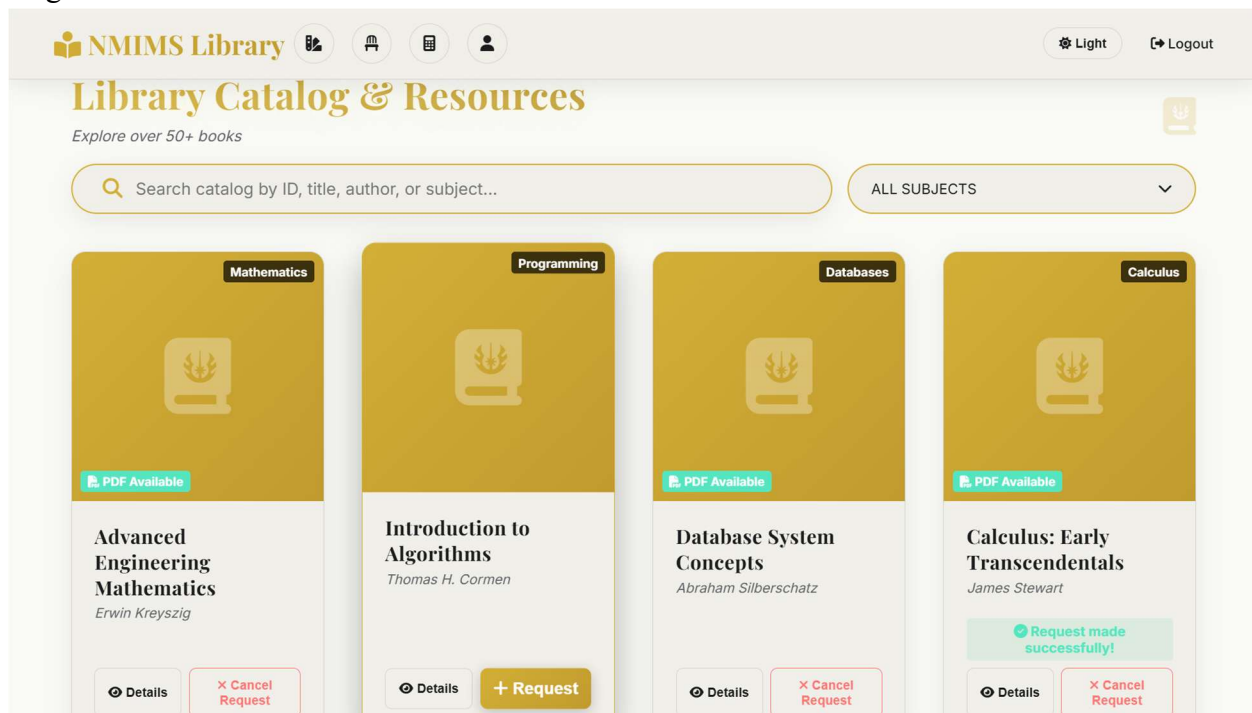
It features an advanced search bar capable of filtering the entire library collection by Title, Author, Subject, or a unique Book ID, allowing for efficient stock updates.

The screenshot shows the 'Global Book Inventory' dashboard. It features a search bar with the placeholder text 'Search inventory by title, author, or ID...' and a dropdown menu for 'All Subjects'. Below the search bar is a table with the following columns: BOOK ID, TITLE, AUTHOR, SUBJECT, and STOCK CONTEXT. The table lists five books with their respective details and stock status.

BOOK ID	TITLE	AUTHOR	SUBJECT	STOCK CONTEXT
#LIB-MA-111	Advanced Engineering Mathematics	Erwin Kreyszig	Mathematics	8 / 10 AVAILABLE / TOTAL
#LIB-PR-112	Introduction to Algorithms	Thomas H. Cormen	Programming	4 / 5 AVAILABLE / TOTAL
#LIB-DA-113	Database System Concepts	Abraham Silberschatz	Databases	8 / 12 AVAILABLE / TOTAL
#LIB-CA-114	Calculus: Early Transcendentals	James Stewart	Calculus	4 / 6 AVAILABLE / TOTAL
#LIB-PH-115	University Physics	Hugh D. Young	Physics	1 / 3 AVAILABLE / TOTAL

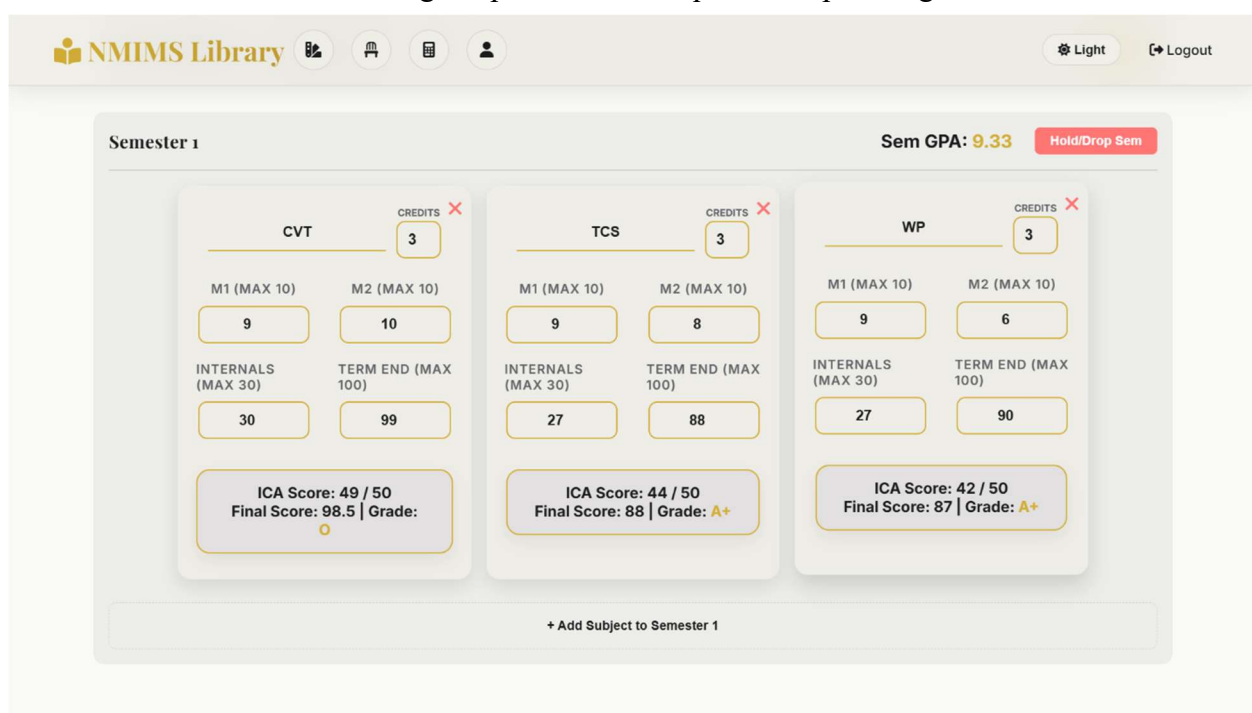
Student Catalog (books.html)

A student-facing inventory exploration module. It provides a clean interface for browsing available textbooks and academic resources, enabling students to submit borrow requests with a single click.



GPA Calculator (gpaCalculator.html)

It includes functionality for recording marks across multiple categories (M1, M2, Internals) and a "Prediction Mode" for estimating the performance required in upcoming exams.



Personal Dashboard (profile.html)

The primary hub for user activity. It features a dashboard grid that balances static user information with a dynamic history of book requests and room bookings, keeping the user informed of their institutional engagement.

 **NMIMS Library**



Light

Logout



John Doe

STUDENT

SAP: 70012345603

Email: john.doe@nmims.in

Course: BTech

Dept: Computer Engineering

Book Requests

Room Bookings

Pending Requests

BOOK	DATE
Advanced Engineering Mathematics	04-11-2026
Organic Chemistry	04-11-2026
Clean Code	04-11-2026
Abstract Algebra	04-11-2026

Accepted Requests

No accepted books.

 **NMIMS Library**



Light

Logout

Pending Requests Dashboard

Pending Book Requests

REQ ID	USER	BOOK TITLE	DATE	ACTION
#56	Ayush Trivedi	Database System Concepts	08-04-2026	ApproveReject
#57	AdminA	Advanced Engineering Mathematics	08-04-2026	ApproveReject
#59	Ayush Trivedi	Organic Chemistry	10-04-2026	ApproveReject
#75	Akshat Vidyarthi	A First Course in Probability	04-11-2026	ApproveReject
#79	Akshat Vidyarthi	Linear Algebra and Its Applications	04-11-2026	ApproveReject
#83	Akshat Vidyarthi	Introduction to Algorithms	04-11-2026	ApproveReject
#84	Akshat Vidyarthi	Advanced Engineering Mathematics	04-11-2026	ApproveReject
#89	Akshat Vidyarthi	Bio-Organic Chemistry	04-11-2026	ApproveReject
#90	Akshat Vidyarthi	Concepts of Physics	04-11-2026	ApproveReject

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Manual Record Entry (addbook.html)

An admin-exclusive page for expanding the library collection. It simplifies the data entry process for new books, ensuring all necessary metadata (Subject, Author, Stock count) is captured correctly.



 Light  Logout

 **Global Book Inventory** 

 **Add New Book to Catalog**

Provision newly acquired physical books into the digital catalog ecosystem.

Book ID (Unique Identifier)

LIB-MA-111

Book Title

Intro to AI

Author

Kevin Miller

Subject

Computer Science 

Copies

1

Add to Library Database

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Observations and Learnings

Throughout the development of the NMIMS Library Management System, several important technical and design insights were gained such as:

Precision in Date Handling:

One of the most critical challenges encountered was managing date and time consistency across different system operations, especially for room bookings and issue/return deadlines. Timezone-related discrepancies initially caused mismatches in booking slots and due dates. This led to the realization that all date operations must be standardized using local time formatting rather than relying on default system or server time.

Scalability of CSS:

The project demonstrated that even without relying on heavy UI frameworks, a visually appealing and scalable interface can be built using pure CSS. By leveraging CSS variables, modular styling, and theme-based design (Dark/Light mode), the system maintains consistency and flexibility. This approach also reduced dependency overhead and improved loading performance.

Real-Time Feedback Mechanisms:

Providing instant visual feedback such as confirmation messages, status indicators (e.g., “Requested”, “Approved”), and dynamic UI updates proved essential in building user trust. Users no longer need to refresh pages or guess system status, which significantly improves usability and perceived responsiveness.

Efficient State Management:

Managing the state of books (available, requested, issued) and rooms (occupied, available) required careful handling to avoid conflicts and inconsistencies. This highlighted the importance of synchronizing frontend UI states with backend data in real time, ensuring that users always interact with accurate information.

Importance of Role-Based Access Control:

Differentiating clearly between admin and student privileges ensured system security and proper functionality. This reinforced the importance of implementing strict access controls in multi-user systems.

Performance Optimization:

Optimizing search functionality and minimizing unnecessary data loads improved system speed. Techniques such as efficient querying and conditional rendering helped maintain smooth performance even with increasing data.

Conclusion

The NMIMS Library Management System successfully demonstrates the transformation of traditional library operations into a modern, efficient, and user-centric digital platform. By integrating key functionalities such as real-time catalog access, streamlined request handling, and interactive room booking, the system significantly reduces manual effort and enhances overall operational efficiency.

The project highlights the importance of combining strong backend logic with an intuitive frontend design to create a seamless user experience for both administrators and students. Features like personalized dashboards, real-time updates, and dual-mode UI (Dark/Light) contribute to improved accessibility, transparency, and user satisfaction.

Furthermore, the system is designed with scalability, security, and performance in mind, making it adaptable to future enhancements and increased user demands. Overall, this project not only addresses the limitations of traditional library systems but also sets a foundation for smarter, technology-driven academic resource management.